

---

# Closing the Frequency Gap: Deploying Vision-Language-Action Models on High-Rate Humanoid Control

---

## Abstract

1 Vision-Language-Action (VLA) models make open-vocabulary robot control  
2 practical, but their inference rate is often far below the joint-control rates that  
3 humanoids need. We treat this mismatch as a deployment bottleneck rather  
4 than a new foundation-model problem. On a DGX Station (Tesla V100), live  
5 UnifoLM-VLA-Base profiling on `G1_Clean_Table` shows a  $\sim 57\times$  gap to 100 Hz  
6 (PyTorch mean **570.7 ms / 1.752 Hz**; Nsight NVTX **593.7 ms / 1.684 Hz**). We  
7 keep the VLA as a sparse language-conditioned intent generator and insert a  
8 fixed Echo State Network (ESN) that turns VLA action commands into smooth  
9 100 Hz joint commands. Only a linear readout is trained with ridge regression. On  
10 `G1_Dex1_Wipe_Table` (200 episodes, train 0–159 / held-out 160–199), the CUDA  
11 ESN (leak  $\alpha=0.3$ , ridge  $\lambda=1$ ) reaches held-out open-loop root-mean-square error  
12 (RMSE)  $2.78 \times 10^{-3} \pm 3.44 \times 10^{-3}$  **rad** ( $\sim 5.6\times$  better than linear upsampling)  
13 and MuJoCo dataset-oracle RMSE  $3.00 \times 10^{-3} \pm 3.56 \times 10^{-3}$  **rad** (grasp/task  
14 success **100%**; mean table contact **74.5%**). Live dual-process control clears 100 Hz  
15 on average (ESN mean 6.92 ms / 145 Hz). With matched `press_table` priors  
16 and motion clamps ( $n=3$  seeds, 30 s), live ESN and zero-order hold (ZOH) both  
17 succeed, and live joint RMSE is near-tied ( $0.521 \pm 0.005$  vs.  $0.532 \pm 0.008$  rad).  
18 The ESN gain that remains clear is offline tracking and jerk, plus the ability to  
19 serve the humanoid control loop while the VLA stays sparse.

## 20 1 Introduction

### 21 1.1 Motivation: why VLAs break on humanoids

22 Recent VLAs such as RT-2 [1], OpenVLA [2], Octo [3], and  $\pi_0$  [4] show that large vision-language  
23 backbones can map instructions to robot actions across many tasks and embodiments [5, 6]. Unitree’s  
24 UnifoLM-VLA [7] brings the same idea to a humanoid stack. That progress is real for semantic  
25 control. It is less clear that the same models are ready for whole-body, contact-rich humanoid  
26 loops [8, 9].

27 The Unitree G1 exposes a **29-degree-of-freedom (DoF)** joint interface that must be served near **100 Hz**  
28 (10 ms) for balance-preserving tracking and safe wiping. Autoregressive and diffusion action heads  
29 often spend hundreds of milliseconds on one inference. If we simply hold each sparse VLA target  
30 with ZOH, the robot sees a staircase joint trajectory for tens of control ticks. That injects high jerk  
31 and can ruin contact even when the high-level plan is correct. We call this the *frequency gap*: the  
32 VLA may be right about *what* to do, yet too slow for *how* the humanoid must move.

33 Two constraints make “just train a faster VLA” hard in many labs. First, the vision-language  
34 backbone already saturates common GPUs, so the control layer should stay cheap and preferably  
35 avoid backpropagation through time (BPTT). Second, classical upsamplers (ZOH, linear interpolation,  
36 joint proportional–integral–derivative (PID) control toward a held setpoint) ignore proprioception  
37 between VLA updates and do not learn demonstration-consistent dynamics.

## 1.2 Approach and scope

We therefore keep UnifoLM as a sparse intent generator ( $\sim 1.7$  Hz on our V100) and insert a leaky Echo State Network (ESN) that produces 100 Hz joint commands. The reservoir weights stay fixed; only a linear readout is fit by ridge regression on public wipe demonstrations (Sec. 4.1). This is a systems paper about deploying an existing humanoid VLA at control rate, not a claim of zero-shot Physical AI, and not a claim of the first VLA temporal upsample.

**Contributions.** To our knowledge, this is the first ridge-trained ESN under a *frozen* humanoid VLA to close a *measured*  $\sim 57\times$  UnifoLM $\rightarrow$ G1 rate gap, with held-out tracking and a disclosed live `press_table` failure—not the first VLA upsample, and not a new ESN algorithm [10, 11].

- We profile UnifoLM-VLA-Base on `G1_Clean_Table` with two instruments and quantify a  $\sim 57$ – $59\times$  gap to 100 Hz.
- We insert a CUDA ESN with input  $u(t) = [q_t; q_{\text{VLA}}^*] \in \mathbb{R}^{58}$  (current joints  $q_t$  and the held VLA target), train only a ridge readout on a declared episode split, and keep a per-task checkpoint library for UnifoLM G1 skills.
- We run dual-process MuJoCo timing that bypasses the Python Global Interpreter Lock (GIL) and compare ESN with ZOH, linear interpolation, and joint PID under live UnifoLM.
- We evaluate the ESN as a MuJoCo dataset-oracle on all 40 held-out wipe episodes.
- We run live UnifoLM wipe trials in simulation with interactive cloth, disclosed geometric priors, matched clamps, and  $n=3$  seeds, including contact collapse without `press_table`.

## 2 Related work

**Generalist robot policies.** Large-scale robot datasets and RT-X models [5] catalyzed open generalist policies such as Octo [3] and OpenVLA [2]. Flow-matching VLAs such as  $\pi_0$  [4] push action generation further, and recent surveys map the remaining gaps for embodied AI [6, 8]. Humanoid-oriented stacks such as UnifoLM [7] and GR00T [9] make the control-rate question concrete: semantic competence is not the same as a 100 Hz whole-body loop.

**Action generation and temporal structure.** Diffusion Policy [12] and action chunking [13] improve smoothness by predicting action sequences rather than single steps. Memory-augmented VLAs explore longer horizon structure [14]. Faster or smaller VLAs such as TinyVLA [15] attack latency inside the policy. We take a complementary route: leave the foundation VLA sparse, and add a light dynamical bridge underneath it. That is not the first way to upsample or smooth VLA actions; it is a training-light alternative that leaves the backbone frozen.

**Reservoir computing for fast control.** Echo State Networks provide trainable-light temporal filtering with fixed recurrent weights [10, 16]. Ihimekpen et al. showed that a fixed reservoir with a ridge-trained readout can identify and control spring-mass systems with switched dynamics [11]. We reuse that training-light property—not a new ESN algorithm—to close the measured UnifoLM-to-G1 rate gap, without backpropagating through the reservoir.

## 3 Method

### 3.1 System architecture

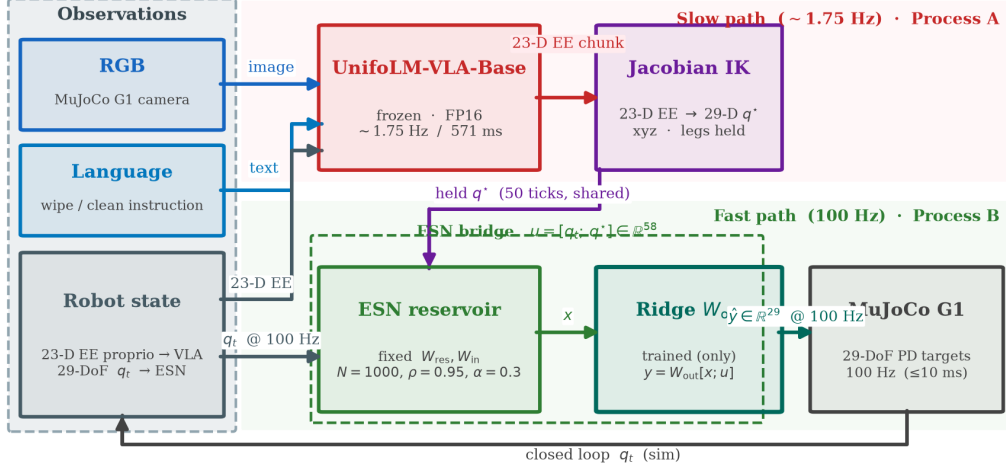
Figure 1 shows the hybrid stack. Frozen UnifoLM emits sparse 23-D end-effector (EE) intents. A Jacobian inverse-kinematics (IK) bridge lifts them to 29-D joint targets and holds  $q^*$  for 50 ticks. A fixed ESN reservoir with a ridge-trained readout then produces 100 Hz MuJoCo PD commands conditioned on current joints.

- **Perception / VLA:** Frozen UnifoLM-VLA-Base consumes RGB, language, and 23-D EE proprioception, and emits 23-D EE action chunks. A Jacobian IK bridge maps EE  $xyz$  to 29-D  $q^*$  (legs held) and latches the target for 50 ticks.
- **ESN bridge:** Leaky reservoir

$$x(t) = (1 - \alpha)x(t - 1) + \alpha \tanh(W_{\text{res}}x(t - 1) + W_{\text{in}}u(t)),$$

## VLA + ESN hybrid control for Unitree G1 (29-DoF, MuJoCo)

Frequency gap: UnifoLM  $\sim 1.75$  Hz  $\rightarrow$  ESN bridge  $\rightarrow$  100 Hz ( $\sim 57\times$ )



**Figure 1:** Two-rate VLA+ESN stack on simulated G1 (29 DoF). Frozen UnifoLM-VLA-Base ( $\sim 1.75$  Hz, V100) emits 23-D EE chunks; Jacobian IK holds  $q^*$  for 50 ticks. Only  $W_{\text{out}}$  is trained. Process B closes the  $\sim 57\times$  gap with 100 Hz MuJoCo PD.

84 with reservoir state  $x(t)$ , leak rate  $\alpha$ , sparse reservoir weights  $W_{\text{res}}$  (size  $N=1000$ , spectral radius  
85  $\rho=0.95$ , sparsity 0.9), and input weights  $W_{\text{in}}$ . The readout  $y(t) = W_{\text{out}}[x(t); u(t)]$  is fit by ridge  
86 regression (penalty  $\lambda$ ) on 100 Hz joints with VLA tokens held for 50 ticks ( $u \in \mathbb{R}^{58}$ ,  $y \in \mathbb{R}^{29}$ ).  
87 We sweep  $\alpha \in \{0.05, 0.1, 0.3\}$  and  $\lambda \in \{10^{-4}, 10^{-2}, 1\}$  and select by an  $\text{RMSE} \times \text{jerk}$  score.

### 88 3.2 Baselines

89 We compare four bridges under the same sparse VLA intents: (1) pure VLA ZOH, (2) VLA + linear  
90 interpolation, (3) VLA + joint PID toward the held target, (4) VLA + ESN (ours).

## 91 4 Experiments

### 92 4.1 Dataset, setup, and metrics

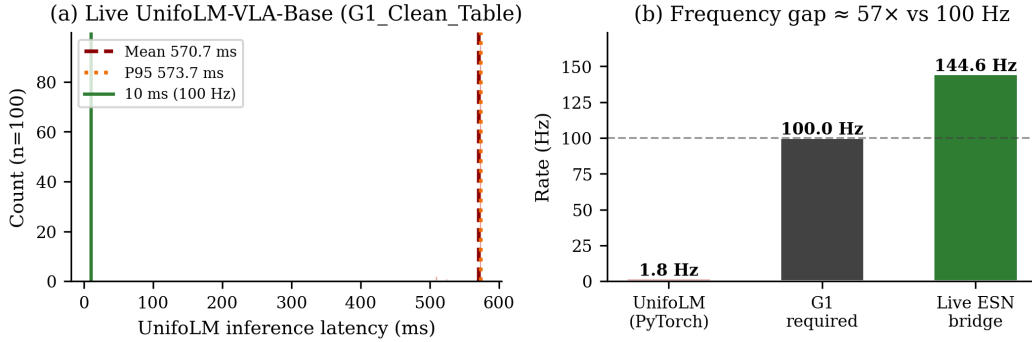
93 ESN training, offline upsample baselines, and MuJoCo wipe-oracle evaluations use  
94 G1\_Dex1\_Wipe\_Table from unitreerobotics (Hugging Face train split): **200** episodes (75,909  
95 frames;  $\sim 0.70$  h), train episodes 0–159 / held-out 160–199. We keep detailed wipe claims inside  
96 this corpus, while the training stack also supports a per-task ESN library over other UnifoLM G1  
97 skills. Figure 3 shows the episode-length distribution and the episode-level 80/20 split. Hardware  
98 is a Unitree G1 (29 DoF) simulated stack on an NVIDIA DGX Station (Tesla V100-DGXS-32GB).  
99 Profiling uses live UnifoLM-VLA-Base on G1 Clean-Table in half precision (FP16). We report task  
100 success (grasp / wipe contact), joint RMSE, trajectory jerk, and control rate.

101 Three evaluation settings are kept separate. Open-loop RMSE compares ESN (or baseline) joint  
102 commands to held-out demonstration joints. The MuJoCo dataset-oracle replays those tokens in  
103 simulation. Live wipe runs UnifoLM and the bridge in MuJoCo with interactive cloth. None of these  
104 is hardware closed-loop control. Reported  $\pm$  values are standard deviations over held-out episodes or  
105 live seeds.

### 106 4.2 Results

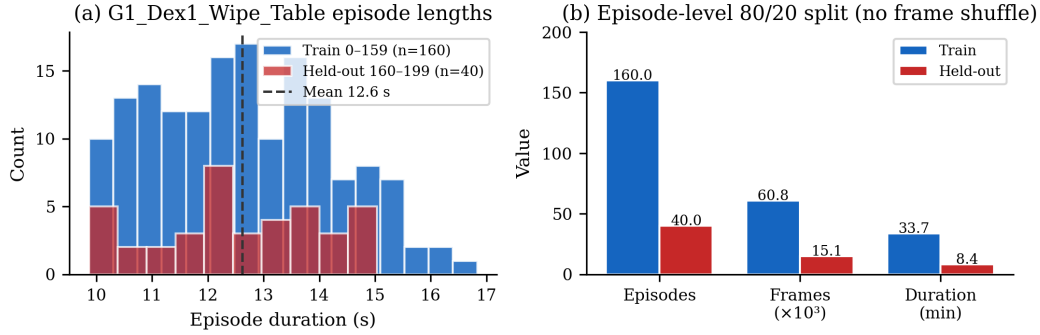
107 Table 1 summarizes the measurements. Table 2 isolates held-out tracking and jerk, which is where  
108 ESN separates from classical upsamplers. Table 3 checks that the same bridge stays in a low open-

### Measured frequency gap on DGX V100



**Figure 2:** Measured UnifoLM frequency gap on DGX V100 (PyTorch  $n=100$  on G1\_Clean\_Table): mean 570.7 ms / 1.752 Hz vs. the 100 Hz G1 budget; live ESN dual-process clears 100 Hz mean.

### Wipe-table corpus: 200 eps, 75,909 frames, $\sim 0.70$ h @ $\sim 30$ Hz



**Figure 3:** Wipe-table corpus used for all ESN / baseline / oracle claims: episode durations and the episode-level 80/20 train (0–159) / held-out (160–199) split on G1\_Dex1\_Wipe\_Table.

**Table 1:** Summary of measured results.

| Measurement                     | Result                                             |
|---------------------------------|----------------------------------------------------|
| VLA latency (PyTorch, $n=100$ ) | 570.7 ms / 1.75 Hz ( $57\times$ below 100 Hz)      |
| VLA latency (Nsight)            | 593.7 ms / 1.68 Hz                                 |
| ESN held-out RMSE               | $2.78 \times 10^{-3} \pm 3.44 \times 10^{-3}$ rad  |
| ESN vs. linear RMSE             | $\sim 5.6\times$ lower                             |
| Live ESN step time              | 6.92 ms / 145 Hz                                   |
| MuJoCo oracle ( $n=40$ )        | 100% grasp; 74.5% contact                          |
| Live wipe RMSE (ESN vs. ZOH)    | $0.521 \pm 0.005$ vs. $0.532 \pm 0.008$ rad (tied) |

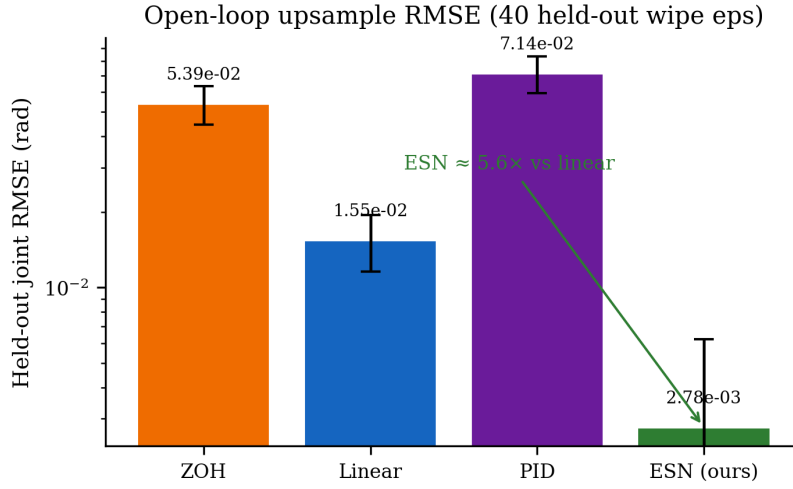
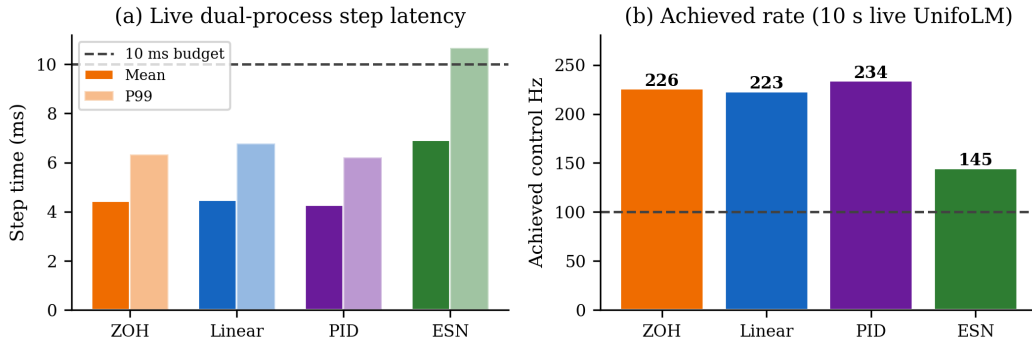
109 loop RMSE regime on two additional UnifoLM G1 skills. Figures 2–6 show the latency gap, dataset  
 110 split, offline baselines, live timing, and the contact ladder.

**Table 2:** Held-out joint tracking on wipe episodes 160–199.

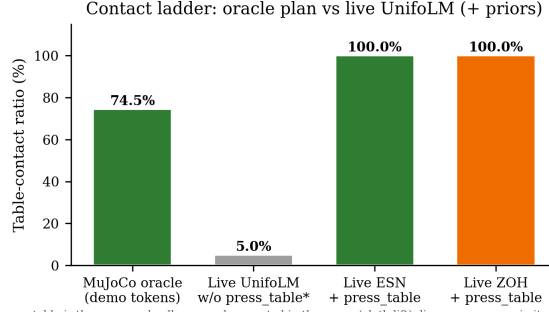
| Method            | RMSE (rad)                                                      | Jerk vs. linear |
|-------------------|-----------------------------------------------------------------|-----------------|
| ZOH               | $5.39 \times 10^{-2} \pm 9.44 \times 10^{-3}$                   | 50×             |
| Linear            | $1.55 \times 10^{-2} \pm 3.93 \times 10^{-3}$                   | 1               |
| PID               | $7.14 \times 10^{-2} \pm 1.19 \times 10^{-2}$                   | 1.43×           |
| <b>ESN (ours)</b> | <b><math>2.78 \times 10^{-3} \pm 3.44 \times 10^{-3}</math></b> | 1.44×           |

**Table 3:** Held-out ESN joint RMSE on three tasks (open-loop tracking).

| Task        | Episodes | RMSE (rad)                                    |
|-------------|----------|-----------------------------------------------|
| Wipe table  | 40       | $2.78 \times 10^{-3} \pm 3.44 \times 10^{-3}$ |
| Clean table | 40       | $3.68 \times 10^{-3} \pm 6.72 \times 10^{-3}$ |
| Stack block | 40       | $4.25 \times 10^{-3} \pm 2.84 \times 10^{-3}$ |

**Figure 4:** Open-loop held-out RMSE on wipe episodes 160–199 (log scale): ESN is  $\sim 5.6\times$  lower than linear upsampling.**GIL-free dual-process MuJoCo + bridge under live UnifoLM****Figure 5:** Live dual-process MuJoCo+bridge under UnifoLM (10 s): mean/P99 step latency and achieved control Hz for ZOH, linear, PID, and ESN.

## 5 Discussion and limitations



\* ~5% without press\_table is the measured collapse mode reported in the paper (cloth lift); live rows use proximately-synthetic Dex1 + press\_table.

**Figure 6:** Contact ladder: oracle demo tokens (74.5%) vs. live UnifoLM without `press_table` (~5% collapse) vs. live ESN/ZOH with a disclosed geometric press prior (100%).

**Frequency gap.** Two independent profilers agree on a  $\sim 57\text{--}59\times$  gap to 100 Hz. Both point to large general matrix multiplies (GEMMs) as the dominant cost; Nsight also highlights host-to-device pressure. That is why we keep the VLA sparse and put a cheap bridge on the joint loop.

**Data scope.** Training and the primary live wipe trials use one task and about 0.7 h of Dex1 demonstrations with an episode-level 80/20 split. Open-loop held-out joint RMSE also stays in the  $10^{-3}$  rad regime on clean-table and stack-block (Table 3), which suggests the bridge is not wipe-only. That comparison is open-loop tracking of demonstration joints, not live multi-task control, and it does *not* show zero-shot humanoid competence.

**What ESN wins, and what stays tied.** Offline, ESN is  $\sim 5.6\times$  better than linear upsampling and much less jerky than ZOH (Table 2). That remains the clearest empirical reason to prefer the reservoir. Live, the dual-process ESN clears 100 Hz on average (145 Hz mean), with GPU-contention tails above 10 ms when UnifoLM shares the device. On interactive cloth, removing the table-plane press prior collapses contact to about 5%. With matched `press_table` priors and motion clamps over  $n=3$  seeds (init eps 160/170/180, 30 s), both ESN and ZOH reach 100% contact and task success, and live joint RMSE is near-tied ( $0.521 \pm 0.005$  vs.  $0.532 \pm 0.008$  rad). So we do not claim a live joint-RMSE win under identical clamps. We do claim a practical deployment pattern: use the VLA for sparse intent, use a light dynamical bridge for rate, and report the geometric priors that still decide contact success.

**Onboard sensing field of view.** Live G1 head-camera bring-up (RGB+depth over ZMQ at  $\sim 10$  Hz) shows a strongly downward ego view: floor and near-field workspace dominate, with little forward horizon. That is a deployment challenge for any vision-conditioned outer loop and is independent of the ESN bridge. Hardware transfer therefore needs camera pitch / mount checks, and possibly wrist or multi-view inputs, before a vision-conditioned wipe can be claimed on the robot.

## 6 Conclusion

We studied the frequency gap between UnifoLM-VLA-Base inference and G1 100 Hz control as a Physical AI deployment problem. To our knowledge, this is the first ridge-trained ESN under a frozen humanoid VLA used to close that *measured*  $\sim 57\times$  UnifoLM $\rightarrow$ G1 gap, with held-out tracking and a disclosed live `press_table` failure. We do not claim the first VLA upsample, a new ESN method, or a live RMSE win: under matched priors and clamps in simulation, ESN and ZOH succeed together. The remaining ESN evidence is offline tracking and jerk, plus a joint loop that stays at 100 Hz while the VLA stays sparse. Evaluating the stack on a physical G1 is the natural next step.

## References

- [1] Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Xi Chen, Karol Choromanski, Tianli Ding, Danny Driess, Avinava Dubey, Chelsea Finn, et al. RT-2: Vision-language-action models transfer web knowledge to robotic control. In *Proceedings of the 7th Conference on Robot Learning (CoRL)*, 2023.

- [2] Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Foster, Grace Lam, Pannag Sanketi, Quan Vuong, Thomas Kollar, Benjamin Burchfiel, Russ Tedrake, Dorsa Sadigh, Sergey Levine, Percy Liang, and Chelsea Finn. OpenVLA: An open-source vision-language-action model. In *Proceedings of the 8th Conference on Robot Learning (CoRL)*, pages 2679–2713, 2025.
- [3] Octo Model Team, Dibya Ghosh, Homer Walke, Karl Pertsch, Kevin Black, Oier Mees, Sudeep Dasari, Joey Hejna, Tobias Kreiman, Charles Xu, et al. Octo: An open-source generalist robot policy. In *Robotics: Science and Systems (RSS)*, 2024.
- [4] Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, et al.  $\pi_0$ : A vision-language-action flow model for general robot control. *arXiv preprint arXiv:2410.24164*, 2024.
- [5] Abby O’Neill, Abdul Rehman, Abhinav Gupta, Abhiram Maddukuri, Abhishek Gupta, Abhishek Padalkar, Abraham Lee, Acorn Pooley, Agrim Gupta, Ajay Mandlekar, et al. Open X-embodiment: Robotic learning datasets and RT-X models. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2024.
- [6] Yuen Ma, Ziyu Song, Yuzheng Zhuang, Jianye Hao, and Irwin King. A survey on vision-language-action models for embodied AI. *arXiv preprint arXiv:2405.14093*, 2024.
- [7] Unitree Robotics. UnifoLM-VLA: Vision-language-action for humanoid manipulation. <https://github.com/unitreerobotics/unifolm-vla>, 2026. Models and datasets on Hugging Face (unitreerobotics).
- [8] Soujanya Poria, Navonil Majumder, Chia-Yu Hung, et al. 10 open challenges steering the future of vision-language-action models. *arXiv preprint arXiv:2511.05936*, 2025.
- [9] NVIDIA et al. GR00T N1: An open foundation model for generalist humanoid robots. *arXiv preprint arXiv:2503.14734*, 2025.
- [10] Herbert Jaeger. The “echo state” approach to analysing and training recurrent neural networks. *GMD Report*, (148), 2001.
- [11] Osemudiamen Andrew Ihimekpen, Xiangfang Li, and Lijun Qian. Modeling and control of spring-mass systems with switched dynamics using reservoir computing. *IEEE Access*, 14:77825–77838, 2026. doi: 10.1109/ACCESS.2026.3693102.
- [12] Cheng Chi, Siyuan Feng, Yilun Du, Zhenjia Xu, Eric Cousineau, Benjamin Burchfiel, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. In *Robotics: Science and Systems (RSS)*, 2023.
- [13] Tony Z. Zhao, Vikash Kumar, Sergey Levine, and Chelsea Finn. Learning fine-grained bimanual manipulation with low-cost hardware. In *Robotics: Science and Systems (RSS)*, 2023. Action Chunking with Transformers (ACT).
- [14] Sanjay Hareesh, Daniel Dijkman, Apratim Bhattacharyya, and Roland Memisevic. Notes-to-self: Scratchpad augmented VLAs for memory dependent manipulation tasks. *arXiv preprint arXiv:2602.21013*, 2026.
- [15] Junjie Wen, Yichen Zhu, Jinming Li, Minjie Zhu, Kun Wu, Zhiyuan Xu, Ning Liu, and Ran Cheng. TinyVLA: Towards fast, data-efficient vision-language-action models for robotic manipulation. In *IEEE Robotics and Automation Letters*, 2025.
- [16] Ruizhi Cao, Chun Guan, Zhongxue Gan, and Siyang Leng. Reviving the dynamics of attacked reservoir computers. *Entropy*, 25(3):515, 2023.